

README: Replication packet for “Rural Roads, Farm Labor Exits, and Cop Fires”

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Overview

The code in this replication package generates the figures and tables shown in the manuscript. The replicator should expect the code to run for about 14 hours.

Data Availability and Provenance Statements

Summary of Availability

- ☐ All data **are** publicly available.
- ☒ Some data **cannot be made** publicly available.
- ☐ **No data can be made** publicly available.

Details on each Data Source

- PMGSY road program, program villages, and village-level data:
Data from the replication package for Rural Roads and Local Economic Development (Asher and Novosad, 2021) (referred to as AN, henceforth). Specifically, two of the data files from AN are used in this analysis: “pmgsy_working_aer_mainsample.dta” and “pmgsy_working_aer.dta” (located in the AN replication package in the folder “data/pmgsy/”).
Village geolocation:
The villages in the PMGSY data from AN above are linked to 2001 census village geocoordinates from a private commercial provider <https://www.mlinfomap.com/map-data.php>. An anonymized version of this linked AN data with geocoordinates used in this study is provided in the replication package with randomly generated village-level identifiers. These files are located in the replication package under “data/raw/an_pmgsy.dta” (corresponding to “pmgsy_working_aer_mainsample.dta” from AN) and “data/raw/pmgsy_runningvar” (corresponding to “pmgsy_working_aer.dta” from AN).
- Fires counts:
Data on fires is from MODIS Collection 6 Hotspot / Active Fire Detections MCD14ML distributed from NASA FIRMS (MODIS). These data are publicly available and were downloaded in shapefile format from <https://firms.modaps.eosdis.nasa.gov/download/> for the India region for the sample period 2001 to 2013. Fire counts, average and total brightness around each sample

village are measured in ArcGIS and the resulting dataset is located in the replication package under “data/raw/pmgysy_fires.dta”.

- Air pollution:

Annual PM_{2.5} measures for villages are constructed from Global Estimates (V4.GL.02) (van Donkelaar et al., 2016) described here and in the cited reference article: <https://sites.wustl.edu/acag/datasets/surface-pm2-5-archive/>. The publicly available data files for Hybrid PM_{2.5} at 0.1° × 0.1° [ug/m³] in netcdf format are downloaded from <https://wustl.box.com/s/h0ev51fbw0iq3quvbdsiyhis8mz0rbc2>. Data are extracted to sample villages using ArcGIS

Monthly emissions are obtained from Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA 2). We use publicly available data from the MERRA-2 tavgM_2d_adg_Nx: 2d, Monthly mean, Time-averaged, Single-Level, Assimilation, Aerosol Diagnostics (extended) V5.12.4 data (GMAO, 2015), downloaded as netcdf files from MDISC, managed by the NASA Goddard Earth Sciences (GES) Data and Information Services Center (DISC) (<https://disc.gsfc.nasa.gov/datasets?project=MERRA-2>). The variables we use are black carbon anthropogenic emissions (BCEMAN), black carbon biomass burning emissions (BCEMBB), organic carbon anthropogenic emissions (OCEMAN), organic carbon biomass burning emissions (OCEMBB). Values from these data are extracted for sample villages using ArcGIS.

The resulting dataset with PM_{2.5} and emissions is located in the replication package under “data/raw/pmgysy_pollution.dta”.

- District-level crop acreages:

District-level data on rice and sugarcane cropped area are downloaded from publicly available ICRISAT District Level Database (ICRISAT DLD, 2020). We use data on crop areas for year 2001 (<http://data.icrisat.org/dld/src/crops.html>) to get share of cropped area under rice and share under sugarcane. Districts in the ICRISAT data are manually matched to the study sample districts and the resulting dataset is located in the replication package under “data/raw/pmgysy_dld.dta”.

- District-level wage rates and returns to education:

Wage rates are obtained from the Employment and Unemployment Survey, July 1999 - June 2000, NSS 55th Round (NSSO, 2000) publicly available here: <https://microdata.gov.in/nada43/index.php/catalog/90>. The datasets used in this study are “Block3-sch10--Household-Characteristics-records.dta” and “Block53-sch10-Persons-daily-activity-time-disposition-Records.dta” (https://microdata.gov.in/nada43/index.php/catalog/90/data_dictionary). District-level wage variables are matched to the study sample districts and the resulting dataset is located under “data/raw/pmgysy_wage.dta”.

District-level returns to education used in heterogeneity analysis for Table A.5. are obtained from the replication package data for "Educational Investment Responses to

Economic Opportunity: Evidence from Indian Road Construction” (Adukia et al., 2020). The dataset “nss_round_55_district.dta” from the Adukia et al., (2020) replication package is used in this analysis. The district-level information is matched to the sample districts in the study, and the resulting dataset is provided under “data/raw/pmgsy_educ.dta”.

- Infant mortality:

Infant mortality data is from the National Family and Health Survey -IV (IIPS, 2016), which is conducted as a part of the global Demographic and Health Survey (DHS). Data is available on request from <https://dhsprogram.com/methodology/survey/survey-display-355.cfm>. The analysis uses the Births Recode dataset (“IAPR74FL.dta”) and Geographic Dataset with the DHS cluster locations (“IAGC72FL.dta”). PM2.5 values from van Donkelaar et al., (2016) are extracted at DHS cluster locations using ArcGIS. Using wind direction at sample villages from ERA 5 (Hersbach et al., 2023), DHS clusters in downwind and other directions are identified using ArcGIS. These data (with cluster identifiers removed) are located under “data/raw/pmgsy_dhs_pm.dta”.

Using the variable b7 from IAPR74FL.dta which shows age at which a child died, we create an indicator for infant mortality that takes value 1 if death occurred before 12 months of age and 0 otherwise (died later or if the child is alive). Using cluster identifiers, infant mortality is matched to DHS locations in downwind and other directions relative to the PMGSY sample villages. The resulting data are provided under “data/raw/pmgsy_birthsdown.dta” and “data/raw/pmgsy_birthsother.dta”.

- District-level agricultural machinery usage:

Agricultural machinery data used in the heterogeneity analysis in Tables A.2. – A.4. is obtained from the Agricultural Input Census (2001) conducted by the Ministry of Agriculture, Government of India (MoA, 2008). The original district-level data was scraped from <http://agcensus.nic.in>. This website has since become unavailable. District-level data matched to the PMGSY sample used in the analysis is added to the replication repository under “data/raw/pmgsy_ag_mech.dta”.

- Village-level stock of agricultural machinery usage, wages, and household machinery use:

Results in Appendix C use restricted-use village and household-level data from the 1999 and 2006 rounds of the Additional Rural Incomes Survey (ARIS)/Rural Economic & Demographic Survey (REDS), administered by the National Council of Applied Economic Research (NCAER). Access details are available here: https://adfdell.pstc.brown.edu/arisreds_data/readme.txt. Anonymized and de-identified data with the select variables used in this analysis are provided with the replication package in “data/raw/appC_data_01.dta” and “data/raw/appC_data_02.dta”.

Dataset list

The script “code/merge_analysis_files.do” creates the main analysis files from the raw data and outputs them in the folder named “data”. The analysis datasets used in the results in the main paper are described in Table 1. Additional datasets provided for replicating the results in the appendices are listed alongside the corresponding code in Table 3 below.

Table 1: Main paper results analysis datasets description

Data file	Source(s)	Notes	Provided
data/maps/fig1_distlayer.rds data/maps/fig1_states.rds	FIRMS	Map data used for Figure 1	Yes
data/gjp_main_working.dta	AN, FIRMS, PM, MERRA-2, DLD, NSSO	Combines multiple data sources, serves as input for Tables 1,2, 3,4, and Figures 2 and 3.	Yes
data/gjp_main_data_birthsdown.dta data/gjp_main_data_birthsother.dta data/gjp_main_data_dhspm50km.dta	AN, PM, DHS, ERA5, FIRMS	Combines multiple data sources, serves as input for Tables 1 and 5, Figure 5.	Yes

Computational requirements

Software Requirements

The scripts used to generate the results are located in the folder named “code”. Figure 1, F.1, and Table H.2. were generated in R (version 4.4.1). All other tables and figures were generated using STATA (version 17.0).

- Stata (code was last run with version 17)
 - estout (version 3.30 25mar2022)
 - binsreg (version 1.2 09-Oct-2022, <https://raw.githubusercontent.com/nppackages/binsreg/master/stata>)
 - reghdfe (version 6.12.2 02Nov2021, installation instructions: <http://scorreia.com/software/reghdfe/install.html>)
 - ivreg2 (version 4.1.11 22Nov2019)

- ivreghdfe (1.1.2 29Sep2022, installation instructions: <https://github.com/sergiocorreia/ivreghdfe#installation>)
 - graph scheme configurations schemepack (v1.4) (installation instructions: <https://github.com/asjadnaqvi/stata-schemepack>) and blindschemes (<https://ideas.repec.org/c/boc/bocode/s458251.html>)
 - coefplot (version 1.8.5 23nov2021)
 - ftools (version 2.49.0 06may2022)
 - program “code/00_setup.do” will define other programs used in the replication and should be run once.
- R 4.4.1
 - pacman (0.5.1)
 - tidyverse (2.0.0)
 - fixest (0.12.1)
 - haven (2.5.4)
 - collapse (0.12.1)
 - sf (0.12.1)
 - tmap (3.3.4)
 - tmaptools (3.1.1)
 - Python 2.7

Memory and Runtime Requirements

Approximate time needed to reproduce the analyses on a standard 2023 desktop machine: 2 - 8 hours. The code was last run on a Apple M1 Pro laptop with MacOS version 12.4 with 16 GB RAM. The replication packet initially requires about 13 GB of hard-drive space.

Instructions to replicators

Scripts and data used for each set of results are shown in the table below. Running “code/master.do” will create the merged analysis files using, create required folders for storing intermediate and final outputs, and run all .do files in sequence (the .R scripts need to be run separately). Figure A.3 and Figure A.5 (in Online Appendix A) use preliminary data from an ongoing study (Jagnani and Mahadevan, 2023). These data will be available after project completion and all

surveys have been finished. Table 2 lists the relevant data files(s) and script that will generate each of the main paper results. Table 3 lists the datasets and scripts used to generate the results in the online appendices.

Table 2: Data files and script to generate main paper results

TABLE/FIGURE	DATA FILE(s)	SCRIPT
Figure 1	maps/fig1_distlayer.rds maps/fig1_states.rds	fig_maps.R
Figure 2	gjp_main_working.dta	main_02.do
Figure 3	gjp_main_working.dta	main_03.do
Figure 4	N.A.	N.A. (schematic)
Figure 5 ¹	gjp_main_data_birthsdown.dta gjp_main_data_birthsother.dta gjp_main_data_dhspm50km.dta	main_04.do
Table 1	gjp_main_working.dta gjp_main_data_birthsdown.dta gjp_main_data_birthsother.dta gjp_main_data_dhspm50km.dta	main_01.do
Table 2	gjp_main_working.dta	main_02.do
Table 3	gjp_main_working.dta	main_02.do
Table 4	gjp_main_working.dta	main_03.do
Table 5	gjp_main_data_birthsdown.dta gjp_main_data_birthsother.dta	main_04.do

¹ Note: The location of the dots on the plots for figure 5 showing the mean within bins of the running variable can perturb slightly each time the script is run. This is due to how stata internally orders data when creating the bins. This does not change the RD linear fit and confidence intervals in the plots nor does it affect the corresponding RD coefficients in Table 5.

Table 3: Data files and script to generate appendix results

Appendix	DATA FILE(s)	Source(s)	Notes	SCRIPT
Appendix A	state_fires.dta	FIRMS	Used for Figure A1.	appendix_A.do
	allindia_fires_pm_early.dta	FIRMS, PM	Combines multiple data sources and used for Figure A2.	
	roads_yearly.dta	AN	Used for Figure A6	
	gjp_main_working.dta	see Table 1	Used for Figures A2, A3, A4, A5, A7, A8, A9 and Tables A1 and A5.	
	labshare_wage_f2001.dta	AN, NSSO, FIRMS	Combines multiple datasets, used for Figure A10.	
Appendix B	pmgsy_runningvar.dta	AN	Used for Figure B1	appendix_B.do
	gjp_main_working.dta	see Table 1	Used for all table in Appendix B	
Appendix C	appC_data_01.dta	AN, ARIS/REDS	Combines PMGSY data with ARIS/REDS village data and used for Tables C1 and C2.	appendix_C.do
	appC_data_02.dta	AN, ARIS/REDS	Combines PMGSY data with ARIS/REDS household data and used for Table C3.	
Appendix D	appD_harvestdates.dta	AN, remotely sensed harvest and	Combines multiple data sources and used as inputs	appendix_D.do

Appendix	DATA FILE(s)	Source(s)	Notes	SCRIPT
	appD_sowdates.dta	sowing dates	for Figures D1 and D2.	
Appendix E	gjp_main_working.dta	see Table 1	Combines multiple data sources, used for Figures E1, E2 and Tables E1, E2.	appendix_E.do
Appendix F	gjp_main_working.dta	see Table 1	Combines multiple data sources, used for Figure F2 and Tables F1, F2, F3.	appendix_F.do
	figF1_distlayer.rds figF1_states.rds	FIRMS, DLD	Map data for Figure F1	fig_maps.R
Appendix G	gjp_main_working.dta	see Table 1	Combines multiple data sources, used for Table G1.	appendix_G.do
Appendix H	gjp_main_working.dta	see Table 1	Data combine multiple datasets, used for Tables H1, H6, and H7	appendix_H.do
	gjp_main_data_birthsdown.dta			
	gjp_main_data_birthsother.dta	see Table 1	Data combine multiple datasets, used for Tables H1 and H3, Figure H1	
	gjp_main_data_dhspm50km.dta			
	appH_matched_data.dta	AN, FIRMS, PM	Data combine multiple datasets, used for Table H2	appendix_H_tabh2.R
	appH_nfhs_wind.dta	NFHS, MERRA	Wind-direction data merged with NFHS and AN, respectively.	
	appH_pmgsy_wind.dta	AN, MERRA	Used for Table H4	appendix_H.do
	appH_2001wdir_births.dta	AN, PM, NFHS, MERRA-2, FIRMS	Data for downwind locations based on 2001 wind direction	
	appH_2001wdir_dhspm			

Appendix	DATA FILE(s)	Source(s)	Notes	SCRIPT
	50km.dta		generated by combining multiple datasets. Used for Table H5.	
	appH_nomovers_dhspm 50km.dta	AN, PM, NFHS, MERRA-2	Data combine multiple datasets, used for Table H6	
	appH_nomigrant_birt hsdown.dta		Data for births and NFHS locations, excluding those born to migrants, generated by combining multiple datasets. Used for Table H8.	
	appH_nomigrant_birt hsother.dta	AN, PM, NFHS, MERRA-2, FIRMS		
	appH_nomigrant_dhsp m50km.dta			

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